

A MULTIVARIATE GAUSS SUPERCONGRUENCE FOR A LEGENDRIAN APÉRY-LIKE SEQUENCE

ANONYMOUS

ABSTRACT. Let

$$G_n = \sum_{k=0}^n 4^k \binom{2n-2k}{n-k}^2 \binom{2k}{k}.$$

Z.-H. Sun conjectured that

$$G_{mp^r} \equiv G_{mp^{r-1}} \pmod{p^{2r}}$$

for every odd prime p and all positive integers m, r . We prove a multivariate strengthening. For

$$P(x, y) = (xy + x - y + 1)(xy - x + y + 1) \\ \times (xy + x + y - 1)(xy - x - y - 1)$$

and $C(n; a, b) = [x^a y^b] P(x, y)^n$, we show that

$$C(np^r; ap^r, bp^r) \equiv C(np^{r-1}; ap^{r-1}, bp^{r-1}) \pmod{p^{2r}}.$$

The proof separates the multinomial expansion into arrays divisible by p and primitive arrays. Two exponent-balance conditions force at least two base- p carries at every relevant digit level in the primitive sector. The divisible sector is paired with the lower-level expansion using the precise Jacobsthal–Kazandzidis congruence. A constant-term representation of G_n makes Sun’s conjecture an immediate corollary. It also gives the one-step congruence $G_N \equiv G_{N/p} \pmod{p^{2v_p(N)}}$ whenever p is an odd prime dividing N .

1. INTRODUCTION

Apéry-like sequences arise as integral solutions of recurrence and differential equations resembling those in Apéry’s proof of the irrationality of $\zeta(2)$ and $\zeta(3)$; see, for example, Zagier’s classification [8]. We consider the second-order Legendrian sequence

$$G_n = \sum_{k=0}^n 4^k \binom{2n-2k}{n-k}^2 \binom{2k}{k} \quad (n \geq 0). \quad (1.1)$$

Its first terms are

$$G_0 = 1, \quad G_1 = 12, \quad G_2 = 164, \quad G_3 = 2352.$$

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In [7, Conjecture 2.5], Z.-H. Sun conjectured that, for every odd prime p and all positive integers m, r ,

$$G_{mp^r} \equiv G_{mp^{r-1}} \pmod{p^{2r}}. \quad (1.2)$$

The modulus in (1.2) gains two powers of p at each dilation. Our argument obtains this gain directly and uniformly, without separating the first layer from the higher layers.

Gorodetsky found a two-variable constant-term representation for (1.1); it appears in Section 5 of the revised arXiv version [2]. After a quadratic substitution, the representing Laurent polynomial factors into four polynomials with coefficients in $\{-1, 0, 1\}$. We show that the resulting multinomial expansion satisfies an order-two Gauss congruence coefficient by coefficient.

The general Dwork congruences for a Laurent polynomial whose Newton polytope has the origin as its unique interior lattice point naturally give congruences of order one; see Mellit and Vlasenko [5]. The extra power at every level in the present problem is therefore not a formal consequence of that theory. Straub [6] proved order-two, and in a subfamily order-three, multivariate supercongruences for a partition-indexed family of rational functions. The rational function $1/(1 - z\Lambda_G(x, y))$ arising here is not among the displayed members of that family, and the balanced four-row coefficient model used below is different; hence the present theorem is not a direct specialization of Straub's result.

Gorodetsky [2, (6.2)] asks whether the Laurent polynomials listed in his Proposition 3.3 satisfy coefficient congruences of order two or three. The second-order Legendrian polynomial in [2, (5.2)] is not one of the polynomials specified in that question. Our result is instead an analogous order-two coefficient congruence for this Legendrian polynomial.

Put

$$\begin{aligned} P(x, y) &= (xy + x - y + 1)(xy - x + y + 1) \\ &\quad \times (xy + x + y - 1)(xy - x - y - 1) \end{aligned} \quad (1.3)$$

and define

$$C(n; a, b) = [x^a y^b] P(x, y)^n \quad (n, a, b \geq 0).$$

For later use, we set $C(n; a, b) = 0$ if $a < 0$ or $b < 0$. Our main result is the following.

Theorem 1.1. *Let p be an odd prime, $r \geq 1$, and $n, a, b \geq 0$. Then*

$$C(np^r; ap^r, bp^r) \equiv C(np^{r-1}; ap^{r-1}, bp^{r-1}) \pmod{p^{2r}}. \quad (1.4)$$

Theorem 1.1 proves more than a diagonal congruence: its Laurent form is precisely the corresponding family of multivariate congruences.

Corollary 1.2. *Z.-H. Sun's congruence (1.2) holds for every odd prime p and all positive integers m, r .*

The proof is elementary and p -adic, relying only on multinomial valuations and the classical Jacobsthal–Kazandzidis congruence; no Dwork-theoretic machinery is required.

2. THE CONSTANT-TERM REPRESENTATION

We first derive the representation directly from (1.1). Set

$$B(t) = \sum_{n \geq 0} \binom{2n}{n}^2 t^n = {}_2F_1 \left(\frac{1}{2}, \frac{1}{2}; 1; 16t \right).$$

The convolution in (1.1) gives

$$\sum_{n \geq 0} G_n t^n = \frac{B(t)}{\sqrt{1-16t}}. \quad (2.1)$$

Pfaff's transformation yields

$$B(t) = \frac{1}{\sqrt{1-16t}} B \left(\frac{-t}{1-16t} \right).$$

Substitution in (2.1) and extraction of the coefficient of t^n give

$$G_n = \sum_{k=0}^n (-1)^k 16^{n-k} \binom{n}{k} \binom{2k}{k}^2. \quad (2.2)$$

Since

$$\text{CT}_x(x+2+x^{-1})^k = \binom{2k}{k},$$

equation (2.2) is equivalent to

$$G_n = \text{CT}_{x,y} \left(16 - (x+2+x^{-1})(y+2+y^{-1}) \right)^n. \quad (2.3)$$

Changing x, y to $-x, -y$ in (2.3) gives the Laurent polynomial recorded in [2, (5.2)]:

$$\Lambda_G(x, y) = (-xy)^{-1} \left(x^2 y^2 - 2x^2 y - 2xy^2 + x^2 - 12xy + y^2 - 2x - 2y + 1 \right). \quad (2.4)$$

Thus

$$G_n = \text{CT} \Lambda_G(x, y)^n.$$

Let

$$A = (x - x^{-1})(y - y^{-1}).$$

A direct calculation gives

$$\Lambda_G(x^2, y^2) = 16 - A^2 = -\frac{P(x, y)}{x^2 y^2},$$

because

$$\begin{aligned} xy(4+A) &= (xy+x-y+1)(xy-x+y+1), \\ xy(4-A) &= -(xy+x+y-1)(xy-x-y-1). \end{aligned}$$

Substitution by x^2, y^2 does not change a constant term. Consequently

$$G_n = (-1)^n [x^{2n} y^{2n}] P(x, y)^n = (-1)^n C(n; 2n, 2n). \quad (2.5)$$

3. BALANCED ARRAYS AND TWO FORCED CARRIES

Fix nonnegative integers N, A, B . Expand each of the four factors in (1.3) to the N th power. For the i th factor let

$$q_i = (a_i, b_i, c_i, d_i), \quad a_i + b_i + c_i + d_i = N,$$

where the four entries record the multiplicities of $xy, x, y, 1$, respectively. A four-row array

$$q = (q_1, q_2, q_3, q_4)$$

contributes to $C(N; A, B)$ precisely when

$$\sum_{i=1}^4 (a_i + b_i) = A, \quad \sum_{i=1}^4 (a_i + c_i) = B. \quad (3.1)$$

We call such an array admissible. The absolute value of its contribution is

$$P_N(q) = \prod_{i=1}^4 \binom{N}{a_i, b_i, c_i, d_i}. \quad (3.2)$$

The sign is a parity character in the entries of q . Explicitly, the four factors in (1.3) contribute the sign exponents

$$c_1, \quad b_2, \quad d_3, \quad b_4 + c_4 + d_4.$$

It follows that multiplication of all entries by an odd integer preserves the sign.

Lemma 3.1. *Let p be a prime, let $R \geq 0$, and suppose that p^R divides each of N, A, B . If q is an admissible array not all of whose entries are divisible by p , then*

$$v_p(P_N(q)) \geq 2R.$$

Proof. The assertion is immediate when $R = 0$, so assume $R \geq 1$. Fix $1 \leq j \leq R$ and put $M = p^j$. Replace each of the sixteen entries of q by its least residue in $[0, M - 1]$. Let the four column totals of these residues be X, Y, Z, W , in the order corresponding to a, b, c, d , and write

$$X + Y + Z + W = MT_j.$$

Each row sum is congruent to N , hence to zero, modulo M , so T_j is a nonnegative integer. It is positive because at least one entry of q is not divisible by p .

We show that $T_j \neq 1$. Suppose otherwise. Then the total residue mass is M . The two balances (3.1) give

$$X + Y \equiv X + Z \equiv 0 \pmod{M}.$$

Both $X + Y$ and $X + Z$ lie between 0 and M , and hence each is either 0 or M . Considering the four possible pairs of values shows that all residue mass must lie in one column:

$$(X, Y, Z, W) \in \{(M, 0, 0, 0), (0, M, 0, 0), (0, 0, M, 0), (0, 0, 0, M)\}.$$

On the other hand, every row residue sum is a multiple of M , and the total of the four row sums is M . Thus exactly one row has residue sum M . Since the mass lies in a single column, one least residue would equal M , a contradiction. Therefore $T_j \geq 2$.

More explicitly, Legendre's formula gives the level- j contribution to the valuation of the product in (3.2) as

$$\sum_{i=1}^4 \left(\left\lfloor \frac{N}{M} \right\rfloor - \sum_{\alpha \in \{a,b,c,d\}} \left\lfloor \frac{q_{i,\alpha}}{M} \right\rfloor \right) = \frac{X + Y + Z + W}{M} = T_j.$$

All contributions for $j > R$ are nonnegative. Hence

$$v_p(P_N(q)) \geq \sum_{j=1}^R T_j \geq 2R.$$

□

Remark 3.2. The lower bound is sharp. When $N = p^R$ and $A = B = 2N$, the rows

$$(0, 0, 0, N), \quad (0, 0, 1, N-1), \quad (N-1, 1, 0, 0), \quad (N, 0, 0, 0)$$

form an admissible array with total valuation $2R$.

4. SCALED MULTINOMIAL COEFFICIENTS

We use the following consequence of the Jacobsthal–Kazandzidis congruence. The quotient congruences in this section are interpreted in $\mathbb{Q}_p$: thus $X \equiv Y \pmod{p^e}$ means $v_p(X - Y) \geq e$.

Lemma 4.1. *Let p be an odd prime, $s \geq 0$, and let $\mathbf{v} = (v_1, \dots, v_t)$ be a weak composition (zero parts are allowed). Then*

$$\frac{\binom{p^{s+1}|\mathbf{v}|}{p^{s+1}v_1, \dots, p^{s+1}v_t}}{\binom{p^s|\mathbf{v}|}{p^sv_1, \dots, p^sv_t}} \equiv 1 \pmod{p^{2s+2}}. \quad (4.1)$$

The numerator and denominator in (4.1) have the same p -adic valuation.

Proof. Decompose each multinomial coefficient into $t-1$ binomial coefficients. Trivial factors produced by zero parts are omitted. For a nontrivial resulting factor, apply the classical congruence

$$\frac{\binom{pn}{pm}}{\binom{n}{m}} \equiv 1 \pmod{p^{v_p(nm(n-m))+3-\delta_{p,3}}}, \quad (4.2)$$

where $\delta_{p,3} = 1$ for $p = 3$ and zero otherwise; see [4], [3, p. 255], or [1, Lemma 4.1]. In the present application, $n, m, n-m$ contain the common factor p^s . The exponent in (4.2) is therefore at least $3s+2$ for $p = 3$ and at least $3s+3$ for $p \geq 5$. Both bounds are at least $2s+2$, so multiplication proves (4.1).

The valuation assertion follows from Legendre's digit-sum formula: multiplication of all arguments by p merely shifts their base- p digits. □

5. PROOF OF THE MULTIVARIATE CONGRUENCE

We prove a slightly stronger one-step statement.

Theorem 5.1. *Let p be an odd prime, and let $N, A, B \geq 0$ be not all zero. Suppose that p divides each of N, A, B , and put*

$$R = \min\{v_p(N), v_p(A), v_p(B)\},$$

with the convention $v_p(0) = +\infty$. Then

$$C(N; A, B) \equiv C(N/p; A/p, B/p) \pmod{p^{2R}}. \quad (5.1)$$

Proof. If $N = 0$, then $A + B > 0$, and both coefficients in (5.1) are zero. We may therefore assume $N > 0$; in particular, $R \geq 1$. Split the admissible arrays contributing to $C(N; A, B)$ into two classes. By Lemma 3.1, every array not wholly divisible by p contributes zero modulo p^{2R} .

Every array in the other class is uniquely $q = pu$, where u is admissible for $C(N/p; A/p, B/p)$. This gives a bijection with all arrays in the lower expansion, and the signs agree because p is odd.

Let

$$s = \min_{i,\alpha} v_p(u_{i,\alpha}),$$

where zero entries have infinite valuation, and write $u = p^s v$. At least one of $N/p, A/p, B/p$ has exact valuation $R - 1$, so $0 \leq s \leq R - 1$. The array v is primitive, and its three parameters are all divisible by p^{R-1-s} . Common scaling does not change a multinomial valuation. Lemma 3.1 therefore gives

$$v_p(P_{N/p}(u)) \geq 2(R - 1 - s). \quad (5.2)$$

Apply Lemma 4.1 to each of the four rows and multiply the four p -adic unit quotients. We obtain

$$\frac{P_N(pu)}{P_{N/p}(u)} \equiv 1 \pmod{p^{2s+2}}. \quad (5.3)$$

Equations (5.2) and (5.3) imply

$$v_p(P_N(pu) - P_{N/p}(u)) \geq 2(R - 1 - s) + 2s + 2 = 2R.$$

Thus every paired divisible term agrees modulo p^{2R} . Adding these terms and the primitive terms proves (5.1). $\square$

Proof of Theorem 1.1. Apply Theorem 5.1 with

$$(N, A, B) = (np^r, ap^r, bp^r).$$

If $n = a = b = 0$, both coefficients in (1.4) are one; otherwise the minimum valuation is at least r . $\square$

6. LAURENT FORM AND SUN'S CONJECTURE

Set

$$\Lambda(x, y) = -\frac{P(x, y)}{x^2 y^2}.$$

For $n \geq 0$ and $h, k \in \mathbb{Z}$, define

$$D(n; h, k) = \text{CT}(\Lambda(x, y)^n x^h y^k).$$

Theorem 1.1 is equivalent to

$$D(np^r; hp^r, kp^r) \equiv D(np^{r-1}; hp^{r-1}, kp^{r-1}) \pmod{p^{2r}}. \quad (6.1)$$

Indeed,

$$D(n; h, k) = (-1)^n C(n; 2n - h, 2n - k),$$

where the convention following the definition of C is used. The two signs agree under dilation by an odd prime. If both ordinary-polynomial indices are nonnegative, Theorem 1.1 applies directly, whether or not the coefficient happens to vanish. If either index is negative, the corresponding index remains negative after dilation, so both sides of (6.1) are zero.

Proof of Corollary 1.2. By (2.5),

$$G_n = (-1)^n C(n; 2n, 2n).$$

Apply Theorem 1.1 with

$$(n, a, b) = (m, 2m, 2m).$$

The signs at levels r and $r - 1$ agree because p is odd, and (1.2) follows. $\square$

Remark 6.1. Theorem 5.1 also yields the stronger one-step form

$$G_N \equiv G_{N/p} \pmod{p^{2v_p(N)}}$$

for every odd prime $p \mid N$.

7. COMPUTATIONAL CHECKS

The proof above is independent of computation. Exact integer programs were nevertheless used to check the transformations and boundary cases. The original convolution (1.1), the alternating formula (2.2), and the signed four-factor expansion were computed through independent code paths. A min-plus enumeration found the sharp minimum $2R$ in five primitive-sector regimes. Finally, all scaled coefficient pairs were checked in the eight regimes

$$(p, r, n) \in \{(3, 1, 1), (3, 1, 2), (3, 2, 1), (3, 2, 2), \\ (5, 1, 1), (5, 1, 2), (5, 2, 1), (7, 1, 1)\}.$$

All 368 coefficient congruences passed. The four exact verification programs used for these checks are supplied in the public repository accompanying this manuscript; they are not part of the logical proof.

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Code availability. The exact verification programs are provided in the public repository accompanying this manuscript.

Use of generative AI. The proof presented in this article was found by OpenAI Codex.

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